

September 6, 2026 10:47 PM GMT

Semiconductor Production Equipment | Japan

SEMICON Taiwan Days 4–5: Takeaways from Advanced Testing Forum and Booth Visits

On Days 4–5 of SEMICON Taiwan 2026, we attended testing-related seminars and visited individual company booths. Below, we summarize key takeaways from the presentations and booth visits.

SEMICON Taiwan 2026, an international semiconductor equipment exhibition, was held from August 31 to September 4 (pre-event forums on August 31–September 1, with the exhibition held September 2–4). On September 3–4, the fourth and fifth days of the event, we attended the Advanced Testing Forum on advanced chip testing and visited individual company booths. The forum featured nine speakers, including KYEC and TSMC, who discussed current testing challenges. Together with our local team, we also conducted booth tours of Kokusai Electric and Lasertec.

Testing was one of the key themes at SEMICON Taiwan 2026. Testing has become a bottleneck for AI/HPC devices, and its technical difficulty is likely to increase further going forward. The exponential rise in test complexity reflects multiple factors acting in combination including: more complex die testing due to higher transistor counts and greater design complexity; more complex package testing due to increasing die counts per package and more interconnects; and more complex memory I/O testing due to higher HBM content and more I/O channels. As AI/HPC devices become increasingly system-level products, component costs rise as devices move through the manufacturing process. This is driving a **shift-left** in testing, with defects screened out more efficiently at earlier stages. Testing is therefore evolving from a simple cost item into a critical manufacturing enabler that helps maximize final product shipments and revenue.

**Charlie Chan covers KYEC and TSMC; Shane Brett covers Teradyne.*

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SEMICONDUCTOR PRODUCTION EQUIPMENT

Japan

Industry View

Attractive

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Advanced Testing Forum

AI devices are creating increasingly complex test challenges, including I/O speeds above 200Gbps, power density exceeding 100W/cm², and associated thermal and warpage issues. Presentations and comments from industry participants indicated that tester supply is already tight and that both the importance of testing and test demand are likely to increase further. Below, we summarize key challenges and development trends in wafer testing, focusing mainly on TSMC's presentation.

Probers

Major prober vendors such as Tokyo Electron and Tokyo Seimitsu (covered by Yoshihito Hasegawa) have recently highlighted rising thermal-control requirements for probers. Thermal management was again a major topic at the seminar.

Thermal challenges: Unlike final test, wafer test lacks a system-level cooling solution. Probe chucks currently have around 16–17 temperature sensors, but unlike final test, they do not directly measure chip temperature. As a result, when a chip heats up rapidly upon power-on, the sensors detect the increase with a lag, creating a large gap between the set temperature and actual junction temperature. Similarly, by the time the tester detects the temperature rise and commands cooling, the chip may already be at a high temperature, leaving the chuck unable to cool it quickly enough. Predictive thermal control synchronized with tester programs has therefore been adopted to compensate for physical response delays.

Die-level testing: Given the difficulty of thermal control at wafer level, die-level testing after singulation is also being considered. Compared with wafer-level testing, this enables more dynamic thermal control as is done in final test to identify Known Good Die/CoW. Tokyo Electron launched its Prexa SDP (Singulated Device Prober) for advanced logic in April 2026, while Advantest announced in December 2025 that it would jointly develop a die-level prober with Tokyo Seimitsu.

Probe Cards

According to TSMC, probe pitch has shrunk from around ~55µm in 2022 to ~30µm currently and is expected to fall below ~25µm by 2030, making µ-bump probing increasingly difficult.

Physical constraints of probe needles: Shortening probe needles is one way to improve electrical performance. Needle length has fallen from around 6mm historically to 3–4mm, and TSMC has already developed 2mm needles. However, very short needles are more prone to tip wear and therefore have shorter lifetimes. Reduced cushioning can also increase pressure on bumps, creating a risk that physical probing itself damages them. Damage to even a single bump can materially affect the overall device, requiring tight yield control. Bumpless structures of hybrid bonding are also difficult to probe directly and require alternative solutions, including test access through RDLs.

High-speed challenges: Design houses have already designed devices with data-transfer speeds above 200Gbps, with speeds above 400Gbps also being seen. As probe needles

become finer, parasitic inductance has a greater impact, disrupting impedance matching during high-speed test-signal transmission and making signal integrity harder to maintain. In addition to shortening needles, development is underway to control impedance by changing the needle material itself.

Thermal challenges: Power density in AI semiconductors is rising sharply generation by generation, increasing fivefold from N5 to N2. Delivering such high power through increasingly fine and densely packed probe needles is causing frequent needle burn, where localized self-heating burns the needles themselves, reducing probe-card lifetime and test reliability. TSMC is seeking to address this through measures including test-program optimization.

Given the limitations of cooling from the prober chuck alone and delays in cooling response, probe cards with enhanced thermal management are also being developed. In addition to conventional cooling from the prober chuck at bottom, these solutions also cool from top through the probe card, directly cooling the device via the probe needles and the top structure.

For chiplet testing, a solution has also been proposed in which active dies are embedded in the probe-card substrate in advance. By combining chiplet dies with active dies mounted on the probe card, inter-chiplet communication and coordinated operation can be tested in mission mode, i.e., under conditions in which the chips are actually integrated and operating together.

Booth Visits

Our booth visits continued to indicate a favorable business environment across equipment suppliers. On equipment pricing, during the April–June earnings season TSMC cited equipment price increases as one factor behind its higher 2026 capex outlook, while equipment makers also cited price increases as one driver of higher WFE forecasts. At SEMICON Taiwan, we confirmed a continued aggressive stance on pricing from equipment suppliers, with similar moves evident among both Japanese and US vendors.

KOKUSAI: The company's large-batch ALD systems, which account for roughly half of sales, have around 50% global share, effectively splitting the market with TEL. Its higher-value-added mini-batch systems face no competition and maintain 100% global share. The company also has around 50% share in single-wafer plasma treatment systems. While mini-batch ALD has yet to secure a POR for leading-edge DRAM, the company expects some adoption in China and is also targeting broader adoption in second-generation GAA logic.

Lasertec: F6/25 was challenging as investment came only from logic customers, but the business environment improved in F6/26 as memory customer investment recovered. For F6/27, Lasertec expects record-high orders, supported by expanding ACTIS demand from both logic and memory customers, with a plan of ¥300–400bn. A200HiT is in the final qualification stage at customer sites, with the first unit expected to ship in the July–September quarter and orders for mass-production applications expected to ramp through 1H27. The firm expects the China sales mix, supported by strong MATRICS demand, to increase in F6/27. Chinese customers are shifting from US competitors to Lasertec amid US export restrictions. However, this warrants caution, as similar shifts occurred in other products such as etchers following tighter US restrictions, which were subsequently

followed by tighter restrictions in Japan.

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	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
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Equal-weight/Hold	1577	43%	395	42%	25%	770	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (09/04/2026)
Suzune Tamura, CFA		
Advantest (6857.T)	O (01/17/2024)	¥33,090
DISCO (6146.T)	O (07/07/2023)	¥54,460
Ebara (6361.T)	O (12/04/2024)	¥4,436
JEOL (6951.T)	E (07/08/2025)	¥6,867
KOKUSAI ELECTRIC (6525.T)	O (12/19/2025)	¥8,295
Lasertec (6920.T)	U (07/28/2025)	¥33,180
Nikon (7731.T)	U (08/08/2023)	¥1,880
SCREEN Holdings (7735.T)	O (11/13/2025)	¥13,030
Tekscend Photomask (429A.T)	E (11/18/2025)	¥3,710
Tokyo Electron (8035.T)	O (12/19/2025)	¥53,320
Ulvac (6728.T)	E (01/17/2024)	¥7,210
Ushio (6925.T)	O (01/05/2026)	¥3,670

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